Online Student-t Processes with an Overall-local Scale Structure for Modelling Non-stationary Data

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Introduction

- Data often comes sequentially in real life.
- Time Series data commonly exhibits non-stationarity and heavy-tailed errors.
- For kernel methods, Student-t Process (TP) can better handle the above issues than Gaussian Process (GP).
- GP and TP however face three challenges:

1. Updating the model in real-time is not trivial.
2. Modelling non-stationarity is challenging for the commonly used stationary kernel functions.
3. Suffering from the cubic computation cost $\mathcal{O}(N^3)$.

- **We introduce a mixture of TPs with an SMC sampler to online infer the non-stationary data in a scalable way.**

Student-t Process for Noisy Data

- Traditional derivation:

$$
\begin{aligned}
\boldsymbol{\Sigma}|\boldsymbol{\theta},h,\nu &\sim IW_N(\nu,\mathbf{K}_{\boldsymbol{\theta}}+|h|\mathbf{I}),\\
\mathbf{y}|\boldsymbol{\Sigma},\nu &\sim \mathcal{N}_N(0,\nu\boldsymbol{\Sigma}),\\
&\int p(\mathbf{y}|\boldsymbol{\Sigma})p(\boldsymbol{\Sigma})d\boldsymbol{\Sigma},\\
\implies \mathbf{y}|\boldsymbol{\theta},h,\nu &\sim \mathcal{T}_N(\nu,0,\mathbf{K}_{\boldsymbol{\theta}}+|h|\mathbf{I}).
\end{aligned} \tag{1}
$$

- Our derivation:

$$
\begin{aligned}
f(\mathbf{X})|\boldsymbol{\theta},\sigma^2 &\sim \mathcal{N}_N(0,\sigma^2\mathbf{K}_{\boldsymbol{\theta}}),\\
\mathbf{y}|f(\mathbf{X}),\sigma^2,h &\sim \mathcal{N}_N(f(\mathbf{X}),\sigma^2|h|\mathbf{I}),\\
\mathbf{y}|\boldsymbol{\theta},\sigma^2,h &= \int P(\mathbf{y}|f(\mathbf{X}),\sigma^2,h)P(f(\mathbf{X})|\boldsymbol{\theta},\sigma^2)df(\mathbf{X}),\\
&\sim \mathcal{N}_N(0,\sigma^2(\mathbf{K}_{\boldsymbol{\theta}}+|h|\mathbf{I})),\\
\sigma^2|\nu &\sim \text{Inv-Gamma}\left(\frac{\nu}{2},\frac{\nu}{2}\right),\\
\mathbf{y}|\boldsymbol{\theta},h,\nu &= \int P(\mathbf{y}|\boldsymbol{\theta},\sigma^2,h)P(\sigma^2|\nu)d\sigma^2\\
&\sim \mathcal{T}_N(\nu,0,\mathbf{K}_{\boldsymbol{\theta}}+|h|\mathbf{I}).
\end{aligned} \tag{2}
$$

- Advantages:

1. Preserving the classical regression model assumption that $y = f(\mathbf{x}) + \sigma\epsilon$, in a $function + noise$ way.
2. TP is used as a prior over the latent function f, the clear prior-likelihood relationship leads to a well-defined tractable marginal likelihood.

Student-t Process Mixture-of-Experts

$$
\begin{aligned}
&\mathbf{x}_i \sim \mathcal{T}_D(\nu_{z_i},\boldsymbol{\mu}_{z_i},\boldsymbol{\Psi}_{z_i}),\\
&\alpha \sim \text{Gamma}(a_0,b_0),\quad z_i|\alpha \sim \text{CRP}(\alpha),\\
&\boldsymbol{\theta}_k \sim \log\mathcal{N}(m_0,s_0^2\mathbf{I}),\quad \nu_k \sim \text{Gamma}(2,0.1),\\
&h_k \sim \mathcal{N}(0,k_0^2),\quad k_0^2 \sim \text{Inv-Gamma}\left(\frac{1}{2},\frac{1}{2}\right),\\
&\mathbf{y}_k|\mathbf{X}_k,\boldsymbol{\theta}_k, \sim \mathcal{T}_{N_k}(\nu_k,0,\mathbf{K}_{\boldsymbol{\theta}_k}+|h_k|\mathbf{I}).
\end{aligned} \tag{3}
$$

- The i−th input $\mathbf{x}_i \in \mathbb{R}^D$ is modelled as a Dirichlet process Gaussian-inverse Wishart mixture model, which marginally gives a multivariate t likelihood.
- The mixture indicator z_i is modelled by the Chinese Restaurant Process (CRP), which allows the number of mixtures to be adaptive.
- Within k−th mixture, the real-valued output $\mathbf{y}_k$ is a TP with parameters $\boldsymbol{\theta}_k, h_k, \nu_k$ following our previous derivation.

When a New Data Comes

1. Assign it to mixture k (set $z_i = k$) according to CRP.
2. Update the input's parameters by the conjugate results between Gaussian and inverse Wishart.
3. Sample the concentration parameter α from its posterior by a variable augmentation scheme.
4. Gibbs sample the degree of freedom ν_k by a variable augmentation scheme based on our derivation:
   - Sample σ_k^2 from its posterior due to the conjugacy between Gaussian and inverse Gamma.
   - Sample ν_k conditional on the newly sampled σ_k^2 by using the slice sampler.
5. Sample k_0^2 from its inverse Gamma posterior.
6. Sample $\boldsymbol{\theta}_k, h_k$ by using elliptical slice sampler.

Sequential Monte Carlo for Online Inference

- Sequential Monte Carlo (SMC) follows importance sampling (IS), where particles $\theta^{(j)}, j = 1,\cdots,J$ are from a proposal distribution $Q(\theta)$ to approximate an integral of function $f(\theta)$ over the target distribution $P(\theta)$:

$$
\int f(\theta)P(\theta)d\theta \approx \sum_{j=1}^{J} w^{(j)}f(\theta^{(j)}),\ \ w^{(j)} = \frac{P(\theta^{(j)})}{Q(\theta^{(j)})} \tag{4}
$$

- Concerning our SMC inference, the initial weight and the updating are as below:

$$
\begin{aligned}
w_1^{(j)} &\propto P(y_1|z_1^{(j)},\mathbf{x}_1,\boldsymbol{\theta}^{(j)},h^{(j)},\nu^{(j)})P(\mathbf{x}_1|z_1^{(j)},\alpha^{(j)})\\
w_i^{(j)} &= w_{i-1}^{(j)}P(\mathbf{x}_i|\alpha^{(j)},z_i^{(j)}) \times \frac{P\left(\mathbf{y}_{1:i}|\mathbf{X}_{1:i},\theta_{k,i}^{(j)},h_k^{(j)},\nu_k^{(j)}\right)}{P\left(\mathbf{y}_{1:i-1}|\mathbf{X}_{1:i-1},\theta_k'^{(j)},h_k'^{(j)},\nu_k'^{(j)}\right)}
\end{aligned} \tag{5}
$$

- To avoid the particle degeneracy problem, if $N_{eff} = \frac{1}{\sum_{j=1}^{J}(w_i^{(j)})^2} < \frac{1}{2}$, we resample the particles and reset them to have equal weights.
- A minibatched approximation is employed for the marginal likelihood:

1. Collect a subsample of size B from the N_k observations in mixture k without replacement.
2. Calculate the likelihood based on the B samples.
3. Upweight the likelihood by N_k/B power.
4. Integrate the upweighted likelihood over the priors.

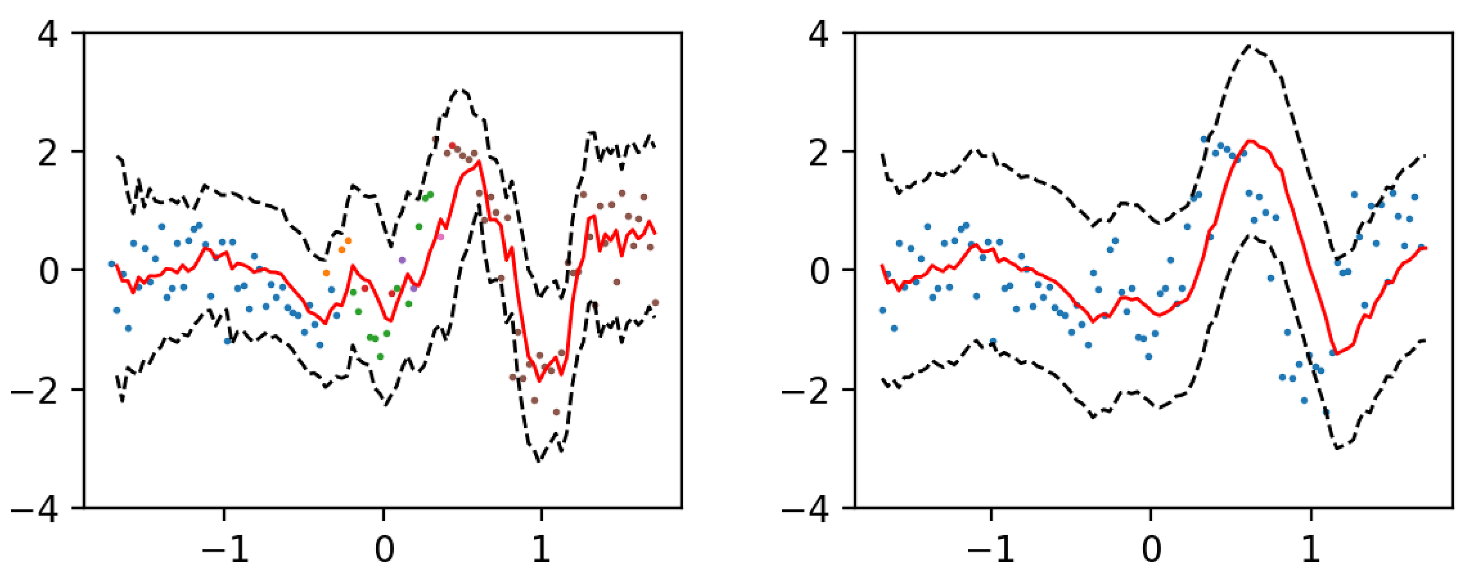
It reduces the overall complexity to $\mathcal{O}(J\min\{N_k,B\}^3/K^2)$. K is the number of mixtures, and B is the minibatch size. The approximation works due to the well-defined marginal likelihood under our derivation.

Experiments

- One-step ahead prediction.
- Results based on three metrics are reported. Predictive mean and 95% prediction interval are plotted.
- Colours indicate the mixture assignment given by the highest-weighted particle.
- Experiments closely related to our intuitions are selected below.

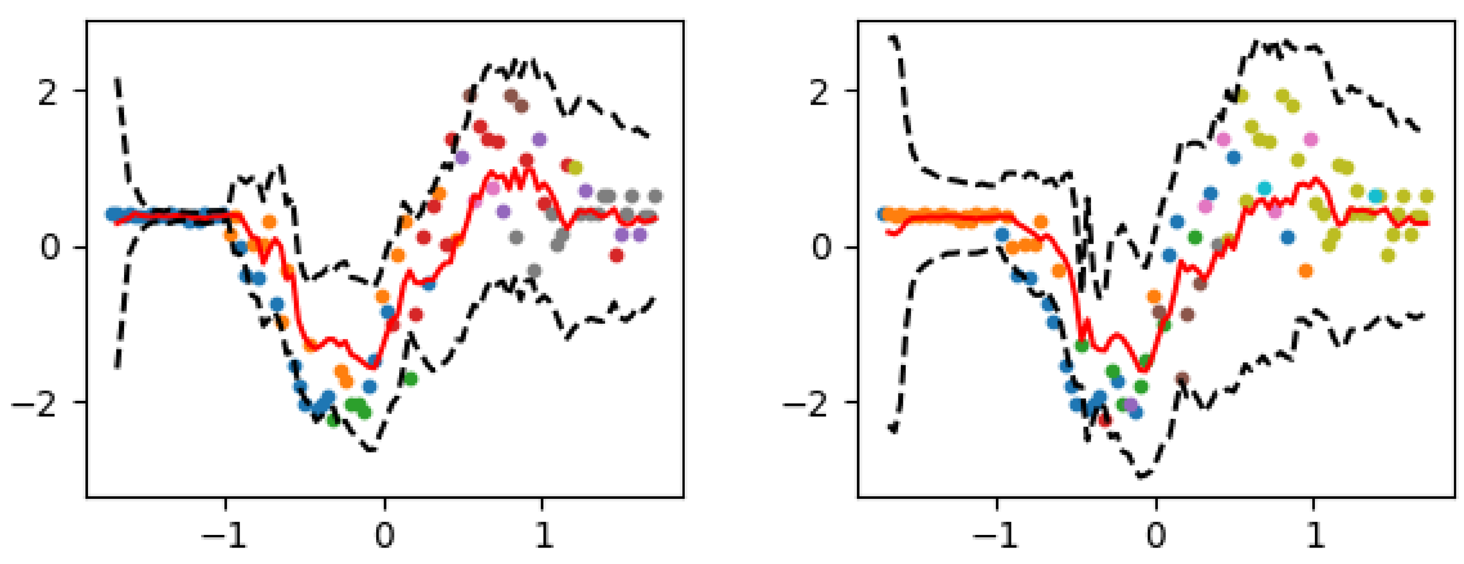
1. Synthetic data by a mixture of TPs (compared with a stationary online GP model WISKI): **MOE gives additional modelling flexibility**.

Synthetic Data	MSE	log likelihood	CRPS
TP-MOE	0.476 (0.008)	-106.678 (2.623)	0.382 (0.003)
WISKI	1.120 (0.000)	-155.905 (0.000)	0.576 (0.000)



2. Real-life data with non-stationary length scale and noise (compared with a GP-MOE model): **TP handles non-stationarity and heavy-tailed errors better than GP**.

Real-life Data	MSE	log likelihood	CRPS
TP-MOE	0.363 (0.028)	-82.468 (13.027)	0.317 (0.014)
GP-MOE	0.381 (0.038)	-88.730 (12.529)	0.324 (0.018)



Conclusion

- Code available at *https://github.com/stlllll/TP-MOE*
- MOE model is especially useful for non-stationary data.
- TP shows superiority over GP in handling non-stationarity and heavy-tailed errors.
- Our derivation of TP enjoys additional modelling flexibility.
- Future work – Extending it in financial prediction tasks.